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SEMICONDUCTOR ELEMENT AND SEMICONDUCTOR DEVICE

Abstract

A semiconductor element includes a main body with a semiconductor layer, an electrode located on one side of the main body in a first direction and electrically conducting to the semiconductor layer, a terminal opposite to the main body with respect to the electrode in the first direction and electrically conducting to the electrode, and a metal bonding layer electrically conducting to the terminal. The terminal includes a first surface facing a side where the electrode is located in the first direction, and an opposite second surface. The bonding layer is opposite to the first surface across the second surface. In a second direction orthogonal to the first direction, the dimension of the periphery of the first surface is not less than 40% and not greater than 60% of the dimension of the periphery of the second surface.

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Background/Summary

FIELD

[0001] The present disclosure relates to a semiconductor element, and a semiconductor device in which the semiconductor element is mounted.

BACKGROUND

[0002] JP-A-2014-165335 discloses an example of a semiconductor element (a semiconductor device in this patent document). The semiconductor element has a substrate having an element formation surface, a pad terminal provided on the element formation surface, a passivation film covering a portion of the pad terminal and the element formation surface, a Cu redistribution wiring extending out from the pad terminal, an organic film covering the Cu redistribution wiring, and a resin film covering the organic film. The surface of the Cu redistribution wiring includes a roughened surface. The organic film is in contact with the roughened surface. With such a configuration, an anchoring effect for the Cu redistribution wiring is generated in the organic film. Furthermore, because the affinity between the organic film and the resin film is relatively high, the adhesion of the resin film to the organic film is enhanced. This suppresses separation of the resin film from the Cu redistribution wiring.

[0003] The semiconductor element disclosed in JP-A-2014-165335 further includes a terminal electrically conducting to the Cu redistribution wiring, and a bonding layer on the terminal. In the semiconductor element, electromigration due to the current density in the terminal affects the bonding layer. If the degree of electromigration is relatively large, there is a risk that cracks will occur in the bonding layer.

Description

DRAWINGS

[0004] FIG. 1 is a plan view of a semiconductor element according to a first embodiment of the present disclosure.

[0005] FIG. 2 is a partial enlarged view of FIG. 1, in which a bonding layer is transparent.

[0006] FIG. 3 is a sectional view taken along line III-III in FIG. 2.

[0007] FIG. 4 is a sectional view taken along line IV-IV in FIG. 2.

[0008] FIG. 5 is a graph for describing the effects of the semiconductor element shown in FIG. 1.

[0009] FIG. 6 is a sectional view of a semiconductor device in which the semiconductor element shown in FIG. 1 is mounted.

[0010] FIG. 7 is a partial enlarged sectional view corresponding to FIG. 3, showing a semiconductor element according to a second embodiment of the present disclosure.

[0011] FIG. 8 is a graph for describing the effects of the semiconductor element shown in FIG. 7.

[0012] FIG. 9 is a partial enlarged sectional view corresponding to FIG. 3, showing a semiconductor element according to a third embodiment of the present disclosure.

[0013] FIG. 10 is a partial enlarged plan view corresponding to FIG. 2, showing a semiconductor element according to a fourth embodiment of the present disclosure.

[0014] FIG. 11 is a sectional view taken along line XI-XI in FIG. 10.

[0015] FIG. 12 is a sectional view of a semiconductor element according to a fifth embodiment of the present disclosure.

EMBODIMENTS

[0016] The details of the present disclosure will be described based on the accompanying drawings.
First Embodiment

[0017] A semiconductor element A10 according to a first embodiment of the present disclosure will be described based on FIGS. 1 to 5. The semiconductor element A10 is, for example, an LSI (Large Scale Integration) referred to as a Wafer Level-Chip Size Package (WL-CSP). The semiconductor element A10 includes a main body 11, a plurality of electrodes 12, a passivation film 20, a first protective film 31, a second protective film 32, a plurality of redistribution wirings 40, a plurality of terminals 50, and a plurality of bonding layers 60. For the convenience of understanding, the bonding layer 60 is transparent in FIG. 2.

[0018] In the description of the semiconductor element A10 and a semiconductor device B that will be described later, the direction that is normal to the obverse surface 11A of the main body 11, described later, is referred to as the “first direction z”. A direction orthogonal to the first direction z is referred to as the “second direction x”. The direction orthogonal to each of the first direction z and the second direction z is referred to as the “third direction y”.

[0019] As shown in FIGS. 3 and 4, the main body 11 includes a semiconductor substrate 111 and a semiconductor layer 112 located on one side of the semiconductor substrate 111 in the first direction z. The main body 11 has an obverse surface 11A facing the above-mentioned one side in the first direction z. The semiconductor layer 112 includes the obverse surface 11A. The semiconductor substrate 111 is made from a silicon wafer, for example. Various semiconductor circuits, such as transistors and diodes, are configured on or near the obverse surface 11A of the semiconductor layer 112.

[0020] As shown in FIGS. 3 and 4, the plurality of electrodes 12 are located on the above-mentioned one side of the main body 11 in the first direction z. The electrodes 12 are in contact with the obverse surface 11A of the main body 11. The electrodes 12 electrically conduct to various semiconductor circuits configured on the semiconductor layer 112. The electrodes 12 contain aluminum (Al), for example. The configurations, such as the arrangement and shapes, of the electrodes 12 in the present disclosure are merely examples. As shown in FIG. 1, the plurality of electrodes 12 include a first electrode 12A, a second electrode 12B, and a third electrode 12C. Each of the first electrode 12A, the second electrode 12B, and the third electrode 12C extends in the third direction y. The first electrode 12A, the second electrode 12B, and the third electrode 12C are arranged side by side along the second direction x. The second electrode 12B is located next to the first electrode 12A in the second direction x. The third electrode 12C is located opposite to the first electrode 12A across the second electrode 12B.

[0021] As shown in FIGS. 3 and 4, the passivation film 20 covers the obverse surface 11A of the main body 11 and a portion of each electrode 12. The passivation film 20 is a thin film containing silicon dioxide (SiO₂) or silicon nitride (Si₃N₄), or a laminate of such thin films.

[0022] As shown in FIGS. 2 to 4, the passivation film 20 has a plurality of openings 21. The openings 21 penetrate in the first direction z. Each of the first electrode 12A, the second electrode 12B, and the third electrode 12C is exposed from one of the openings 21.

[0023] As shown in FIGS. 3 and 4, the second protective film 32 is located between the main body 11 and the first protective film 31 in the first direction z. The second protective film 32 covers a portion of each electrode 12, and the passivation film 20. The second protective film 32 is an insulator containing an organic compound. The second protective film 32 is made of a material containing polyimide. The second protective film 32 includes portions located between the passivation film 20 and the plurality of redistribution wirings 40. As shown in FIG. 3, the second protective film 32 has second openings 321 penetrating in the first direction z. As viewed in the

first direction z, each second opening 321 overlaps with one of the openings 21 of the passivation film 20. In the semiconductor element A10, each second opening 321 exposes one of the electrodes 12. In the semiconductor element A10, one of the second openings 321 exposes the first electrode 12A.

[0024] As shown in FIGS. 3 and 4, the redistribution wirings 40 are located between the electrodes 12 and the terminals 50 in the first direction z. Each of the redistribution wirings 40 electrically conducts to one of the electrodes 12. Each of the redistribution wirings 40 includes a barrier layer that is in contact with at least one of the electrodes 12 and the second protective film 32, a seed layer deposited on the barrier layer, and a plating layer deposited on the seed layer. The barrier layer contains titanium (Ti). Each of the seed layer and the plating layer contains copper (Cu).

[0025] As shown in FIGS. 3 and 4, at least one of the redistribution wirings 40 has a main portion 41, a contact portion 42, and a dented portion 43. The contact portion 42 is electrically connected to one of the electrodes 12. In the semiconductor element A10, the contact portion 42 is electrically connected to the first electrode 12A. The contact portion 42 is in contact with the second protective film 32 and housed in a second opening 321 of the second protective film 32. As viewed in the first direction z, the entirety of the contact portion 42 overlaps with one of the openings 21 of the passivation film 20. The main portion 41 is located opposite to the electrodes 12 across the contact portion 42 in the first direction z. The contact portion 42 is connected to the main portion 41. The main portion 41 is sandwiched between the first protective film 31 and the second protective film 32. The dented portion 43 is recessed in the first direction z from the side of the main portion 41 that faces the first protective film 31 in the first direction z. As viewed in the first direction z, the dented portion 43 overlaps with the contact portion 42. The first protective film 31 is received in the dented portion 43.

[0026] As shown in FIGS. 3 and 4, the first protective film 31 covers the second protective film 32 and the redistribution wirings 40. The first protective film 31 is an insulator containing an organic compound. The first protective film 31 is made of a material containing polyimide. In the semiconductor element A10, the composition of the first protective film 31 is the same as that of the second protective film 32. The first protective film 31 is in contact with the terminals 50. The dimension in the first direction z of the first protective film 31 is greater than the dimension in the first direction z of the second protective film 32. The first protective film 31 has first openings 311 penetrating in the first direction z. Each first opening 311 exposes the main portion 41 of one of the redistribution wirings 40. A portion of one of the terminals 50 is housed in each first opening 311.

[0027] As shown in FIGS. 3 and 4, each of the terminals 50 is located opposite to the main body 11 with respect to the electrodes 12 and the redistribution wirings 40 in the first direction z. Each of the terminals 50 is electrically connected to the main portion 41 of one of the redistribution wirings 40. Thus, each of the terminals 50 electrically conducts to one of the electrodes 12. The terminals 50 contain copper. As shown in FIG. 1, in the semiconductor element A10, the plurality of terminals 50 are arranged in a grid pattern as viewed in the first direction Z.

[0028] As shown in FIGS. 3 and 4, each of the terminals 50 has a first portion 51 and a second portion 52. The first portion 51 is housed in a first opening 311 of the first protective film 31. The second portion 52 is connected to the first portion 51 and protrudes from the first opening 311. As shown in FIG. 2, as viewed in the first direction z, the second portion 52 is located outside the first opening 311.

[0029] As shown in FIGS. 3 and 4, the first portion 51 has a first surface 511 and a first peripheral surface 512. The first surface 511 faces the side where the electrodes 12 and the redistribution wirings 40 are located in the first direction z. The first surface 511 is electrically connected to the main portion 41 of one of the redistribution wirings 40. The first peripheral surface 512 surrounds the first surface 511 as viewed in the first direction z. The first peripheral surface 512 is in contact with the first protective film 31. The first peripheral surface 512 is inclined with respect to each of the second direction x and the third direction y. The cross-sectional area of the first portion 51 in a

direction orthogonal to the first direction z decreases from the second portion **52** toward the first surface **511**.

[0030] As shown in FIGS. **3** and **4**, the second portion **52** has a second surface **521** and a second peripheral surface **522**. The second surface **521** faces away from the first surface **511** of the first portion **51** in the first direction z. The second surface **521** is covered with one of the bonding layers **60**. The second peripheral surface **522** surrounds the second surface **521** as viewed in the first direction z. The second peripheral surface **522** is exposed to the outside of the semiconductor element **A10**. The second peripheral surface **522** is inclined with respect to each of the second direction x and the third direction y. The cross-sectional area of the second portion **52** in a direction orthogonal to the first direction z decreases from the second surface **521** toward the first portion **51**. As viewed in the first direction z, the entirety of the first peripheral surface **512** of the first portion **51** is located inwardly of the second peripheral surface **522**.

[0031] As shown in FIGS. **3** and **4**, in each of the second direction x and the third direction y, the dimension D1 of the periphery of the first surface **511** of the first portion **51** is smaller than the dimension D2 of the periphery of the second surface **521** of the second portion **52**. The dimension D1 is equal to or greater than 40% and equal to or less than 60% of the dimension D2.

[0032] As shown in FIGS. **2** to **4**, each of the terminals **50** has a first recess **53** recessed from the second surface **521** of the second portion **52**. A portion of one of the bonding layers **60** is housed in the first recess **53**.

[0033] As shown in FIG. **3**, each of the bonding layers **60** is located opposite to the first surface **511** across the second surface **521** of one of the terminals **50** in the first direction z. The bonding layers **60** electrically conduct to the terminals **50**, respectively. The bonding layers **60** contain a metal. The bonding layers **60** are solder. Therefore, the composition of the bonding layers **60** includes tin. The melting point of the bonding layers **60** is lower than the melting point of the terminals **50**.

[0034] Next, the semiconductor device B in which the semiconductor element **A10** is mounted will be described based on FIG. **6**.

[0035] As shown in FIG. **6**, the semiconductor device B includes the semiconductor element **A10** and a base **80**. The semiconductor element **A10** is mounted on the base **80**. The base **80** includes a substrate **81** and conductive parts **82**. The substrate **81** is an insulator. The conductive parts **82** contain copper, for example. Each of the terminals **50** of the semiconductor element **A10** is conductively bonded to a conductive part **82** via one of the bonding layers **60**. Thus, the electrodes **12** of the semiconductor elements **A10** electrically conduct to the conductive parts **82**.

[0036] In the semiconductor device B, the base **80** is a wiring board. Alternatively, the base **80** may include only the conductive parts **82** that are leads. The base **80** may include an external terminal located opposite to the conductive parts **82** with respect to the substrate **81** in the first direction z. In the semiconductor device B, the semiconductor element **A10** may be covered with scaling resin such as underfill.

[0037] Next, the effects of the semiconductor element **A10** will be described.

[0038] The semiconductor element **A10** includes the main body **11**, the electrode **12**, the terminal **50**, and the bonding layer **60**. The bonding layer **60** contains a metal. The terminal **50** has the first surface **511** and the second surface **521**. The bonding layer **60** is located opposite to the first surface **511** across the second surface **521**. In the second direction x, the dimension D1 of the periphery of the first surface **511** is equal to or greater than 40% and equal to or less than 60% of the dimension D2 of the periphery of the second surface **521**. Referring now to FIG. **5**, the figure shows the change in the maximum current density at the terminal **50**. The horizontal axis of FIG. **5** shows the ratio [%] of the dimension D1 to the dimension D2. The vertical axis of FIG. **5** shows the maximum current density [A/m.sup.2] at the terminal **50**, where the unit length is along the first direction z. According to FIG. **5**, the inventor has confirmed that the maximum current density at the terminal **50** is relatively low when the ratio of the dimension D1 to the dimension D2 is equal to or greater than 40% and equal to or less than 60%. That is, the present configuration reduces the

maximum current density in the terminal **50**, and hence suppresses the occurrence of cracks in the bonding layer **60** due to electromigration. Therefore, according to the present configuration, it is possible to suppress the occurrence of cracks in the bonding layer **60** due to electromigration in the semiconductor element **A10**.

[0039] The semiconductor element **A10** also includes the redistribution wiring **40** and the first protective film **31**. The first protective film **31** has the first opening **311** penetrating in the first direction **z** and exposing the redistribution wiring **40**. The terminal **50** has the first portion **51** including the first surface **511** and housed in the first opening **311**, and the second portion **52** including the second surface **521** and protruding from the first opening **311**. As viewed in the first direction **z**, the second portion **52** is located outside the first opening **311**. With such a configuration, the redistribution wiring **40** and the first portion **51** are protected by the first protective film **31**, and excessive changes in the current density at the terminal **50** can be suppressed.

[0040] The first portion **51** has the first peripheral surface **512** surrounding the first surface **511** as viewed in the first direction **z**. The first peripheral surface **512** is inclined with respect to the second direction **x**. The cross-sectional area of the first portion **51** in a direction orthogonal to the first direction **z** decreases from the second portion **52** toward the first surface **511**. With such a configuration, the distribution of current density in the first portion **51** can be made uniform.

[0041] The second portion **52** has the second peripheral surface **522** surrounding the second surface **521** as viewed in the first direction **z**. The second peripheral surface **522** is inclined with respect to the second direction **x**. The cross-sectional area of the second portion **52** in a direction orthogonal to the first direction **z** decreases from the second surface **521** toward the first portion **51**. With such a configuration, the distribution of current density in the second portion **52** can be made uniform.

[0042] As viewed in the first direction **z**, the entirety of the first peripheral surface **512** of the first portion **51** is located inwardly of the second peripheral surface **522** of the second portion **52**. With such a configuration, the distribution of current density at the interface between the first portion **51** and the second portion **52** can be made uniform.

Second Embodiment

[0043] A semiconductor element **A20** according to a second embodiment of the present disclosure will be described based on FIGS. **7** and **8**. In these figures, the elements that are identical or similar to those of the semiconductor element **A10** described above are denoted by the same reference signs, and the descriptions thereof are omitted. Incidentally, FIG. **7** corresponds to FIG. **3** that shows the semiconductor element **A10**.

[0044] The semiconductor element **A20** differs from the semiconductor element **A10** in the configuration of the terminals **50**.

[0045] As shown in FIG. **7**, in each of the terminals **50**, the dimension **t2** of the second portion **52** in the first direction **z** is greater than the dimension **t1** of the first portion **51** in the first direction **z**. The dimension **t2** is equal to or greater than 5% of the dimension **D2** in each of the second direction **x** and the third direction **y** of the periphery of the second surface **521** of the second portion **52**.

[0046] Next, the effects of the semiconductor element **A20** will be described.

[0047] The semiconductor element **A20** includes the main body **11**, the electrode **12**, the terminal **50**, and the bonding layer **60**. The bonding layer **60** contains a metal. The terminal **50** has the first surface **511** and the second surface **521**. The bonding layer **60** is located opposite to the first surface **511** across the second surface **521**. In the second direction **x**, the dimension **D1** of the periphery of the first surface **511** is equal to or greater than 40% and equal to or less than 60% of the dimension **D2** of the periphery of the second surface **521**. With such a configuration, it is possible to suppress the occurrence of cracks in the bonding layer **60** due to electromigration in the semiconductor element **A20** as well. The semiconductor element **A20** has a configuration in common with the semiconductor element **A10**, thereby achieving the same effect as the semiconductor element **A10**.

[0048] In the terminals **50** of the semiconductor element **A20**, the dimension $t2$ of the second portion **52** in the first direction z is greater than the dimension $t1$ of the first portion **51** in the first direction z . The dimension $t2$ is equal to or greater than 5% of the dimension $D2$ in the second direction x of the periphery of the second surface **521** of the second portion **52**. Referring now to FIG. **8**, the figure shows the change in the maximum current density at the terminal **50**. The horizontal axis of FIG. **8** shows the ratio [%] of the dimension $t2$ to the dimension $D2$. The vertical axis of FIG. **8** shows the maximum current density [A/mm.sup.2] at the terminal **50**, where the unit length is along the first direction z . According to FIG. **8**, the inventor has confirmed that the maximum current density at the terminal **50** is relatively low when the ratio of dimension $t2$ to dimension $D2$ exceeds 4%. That is, the present configuration reduces the maximum current density in the terminal **50**, and hence suppresses the occurrence of cracks in the bonding layer **60** due to electromigration.

Third Embodiment

[0049] A semiconductor element **A30** according to a third embodiment of the present disclosure will be described based on FIG. **9**. In these figures, the elements that are identical or similar to those of the semiconductor element **A10** described above are denoted by the same reference signs, and the descriptions thereof are omitted. Incidentally, FIG. **9** corresponds to FIG. **3** that shows the semiconductor element **A10**.

[0050] The semiconductor element **A30** differs from the semiconductor element **A20** in that the semiconductor element **A30** further includes intermediate layers **70**.

[0051] As shown in FIG. **9**, each intermediate layer **70** is located between the second surface **521** of one of the terminals **50** and one of the bonding layers **60**. Each intermediate layer **70** is in contact with a second surface **521**, the surface of one of the terminals **50** that defines the first recess **53**, and one of the bonding layers **60**. The intermediate layers **70** contain a metal. The composition of the intermediate layers **70** includes nickel (Ni).

[0052] Next, the effects of the semiconductor element **A30** will be described.

[0053] The semiconductor element **A30** includes the main body **11**, the electrode **12**, the terminal **50**, and the bonding layer **60**. The bonding layer **60** contains a metal. The terminal **50** has the first surface **511** and the second surface **521**. The bonding layer **60** is located opposite to the first surface **511** across the second surface **521**. In the second direction x , the dimension $D1$ of the periphery of the first surface **511** is equal to or greater than 40% and equal to or less than 60% of the dimension $D2$ of the periphery of the second surface **521**. With such a configuration, it is possible to suppress the occurrence of cracks in the bonding layer **60** due to electromigration in the semiconductor element **A30** as well. The semiconductor element **A30** has a configuration in common with the semiconductor element **A10**, thereby achieving the same effect as the semiconductor element **A10**.

[0054] The semiconductor element **A30** further includes the intermediate layer **70** located between the second surface **521** of the terminal **50** and the bonding layer **60**. The composition of the intermediate layer **70** includes nickel. The intermediate layer **70** is in contact with each of the second surface **521** and the bonding layer **60**. With such a configuration, when the semiconductor element **A30** is mounted on the base **80** shown in FIG. **6**, eutectic metal is formed by the intermediate layer **70** and the molten bonding layer **60**. The eutectic metal increases the tensile strength of the bonding layer **60**. This effectively suppresses the occurrence of cracks in the bonding layer **60** due to electromigration.

Fourth Embodiment

[0055] A semiconductor element **A40** according to a fourth embodiment of the present disclosure will be described based on FIGS. **10** and **11**. In these figures, the elements that are identical or similar to those of the semiconductor element **A10** described above are denoted by the same reference signs, and the descriptions thereof are omitted. For the convenience of understanding, the bonding layer **60** is transparent in FIG. **10**. Incidentally, FIG. **10** corresponds to FIG. **2** that shows the semiconductor element **A10**.

[0056] The semiconductor element **A40** differs from the semiconductor element **A30** in the configurations of the second protective film **32**, the redistribution wirings **40**, and the terminals **50**. [0057] As shown in FIGS. **10** and **11**, one of the second openings **321** of the second protective film **32** exposes the second electrode **12B**. As viewed in the first direction **z**, the second opening **321** overlaps with one of the openings **21** of the passivation film **20** that exposes the second electrode **12B**. The contact portion **42** of one of the redistribution wirings **40** is electrically connected to the second electrode **12B**. The contact portion **42** is housed in the second opening **321**.

[0058] As shown in FIG. **11**, the first portion **51** of each of the terminals **50** is received in the dented portion **43** of one of the redistribution wirings **40**. Each of the terminals **50** has a second recess **54**. The second recess **54** is recessed in the first direction **z** from the surface of the terminal **50** that defines the first recess **53**. A portion of one of the bonding layers **60** is housed in the second recess **54**. The intermediate layer **70** is in contact with the surface of the terminal **50** that defines the second recess **54**. As shown in FIG. **10**, as viewed in the first direction **z**, the second recess **54** is located inwardly of the second opening **321** of the second protective film **32**.

[0059] Next, the effects of the semiconductor element **A40** will be described.

[0060] The semiconductor element **A40** includes the main body **11**, the electrode **12**, the terminal **50**, and the bonding layer **60**. The bonding layer **60** contains a metal. The terminal **50** has the first surface **511** and the second surface **521**. The bonding layer **60** is located opposite to the first surface **511** across the second surface **521**. In the second direction **x**, the dimension **D1** of the periphery of the first surface **511** is equal to or greater than 40% and equal to or less than 60% of the dimension **D2** of the periphery of the second surface **521**. With such a configuration, it is possible to suppress the occurrence of cracks in the bonding layers **60** due to electromigration in the semiconductor element **A40** as well. The semiconductor element **A40** has a configuration in common with the semiconductor element **A10**, thereby achieving the same effect as the semiconductor element **A10**.

[0061] In the semiconductor element **A40**, each terminal **50** has a second recess **54** recessed in the first direction **z** from the surface defining the first recess **53**. A portion of a bonding layer **60** is housed in the second recess **54**. Such a configuration enhances the anchoring effect of the bonding layer **60** for the terminal **50**, so that the bonding strength of the bonding layer **60** to the terminal **50** is increased.

[0062] In the semiconductor element **A40**, the first portion **51** of the terminal **50** is received in the dented portion **43** of the redistribution wiring **40**. Such a configuration reduces the electrical resistance at the interface between the redistribution wiring **40** and the terminal **50**.

Fifth Embodiment

[0063] A semiconductor element **A50** according to a fifth embodiment of the present disclosure will be described based on FIG. **12**. In these figures, the elements that are identical or similar to those of the semiconductor element **A10** described above are denoted by the same reference signs, and the descriptions thereof are omitted. Incidentally, FIG. **12** corresponds to FIG. **6** that shows the semiconductor element **A10** constituting the semiconductor device **B**.

[0064] The semiconductor element **A50** differs from the semiconductor element **A10** in that the semiconductor element **A50** further includes a third protective film **33**.

[0065] As shown in FIG. **12**, the third protective film **33** is located opposite to the electrodes **12** across the main body **11** in the first direction **z**. The third protective film **33** covers one side of the main body **11** in the first direction **z**. The third protective film **33** is made of a material containing, for example, an epoxy resin.

[0066] Next, the effects of the semiconductor element **A50** will be described.

[0067] The semiconductor element **A50** includes the main body **11**, the electrode **12**, the terminal **50**, and the bonding layer **60**. The bonding layer **60** contains a metal. The terminal **50** has the first surface **511** and the second surface **521**. The bonding layer **60** is located opposite to the first surface **511** across the second surface **521**. In the second direction **x**, the dimension **D1** of the periphery of the first surface **511** is equal to or greater than 40% and equal to or less than 60% of the dimension

D2 of the periphery of the second surface **521**. With such a configuration, it is possible to suppress the occurrence of cracks in the bonding layers **60** due to electromigration in the semiconductor element **A50** as well. The semiconductor element **A50** has a configuration in common with the semiconductor element **A10**, thereby achieving the same effect as the semiconductor element **A10**. [0068] The semiconductor element **A50** further includes a third protective film **33** located opposite to the electrodes **12** across the main body **11** in the first direction **z**, and the third protective film **33** covers one side of the main body **11** in the first direction **z**. With such a configuration, warping of the main body **11** in the first direction **z** can be reduced during the manufacture of the semiconductor element **A50**.

[0069] The present disclosure is not limited to the above-described embodiments. Various modification in design may be made freely in the specific structure of each part of the present disclosure.

[0070] The present disclosure includes the embodiments described in the following clauses.

Clause 1.

[0071] A semiconductor element (**A10**) comprising: [0072] a main body (**11**) including a semiconductor layer (**112**); [0073] an electrode (**12**) located on one side of the main body (**11**) in a first direction and electrically conducting to the semiconductor layer (**112**); [0074] a terminal (**50**) located opposite to the main body (**11**) with respect to the electrode (**12**) in the first direction and electrically conducting to the electrode (**12**); and [0075] a bonding layer (**60**) electrically conducting to the terminal (**50**), wherein [0076] the bonding layer (**60**) contains a metal, [0077] the terminal (**50**) includes a first surface (**511**) facing a side where the electrode (**12**) is located in the first direction, and a second surface (**521**) facing away from the first surface (**511**) in the first direction, [0078] the bonding layer (**60**) is located opposite to the first surface (**511**) across the second surface (**521**), and [0079] in a second direction orthogonal to the first direction, a dimension (D1) of a periphery of the first surface (**511**) is equal to or greater than 40% and equal to or less than 60% of a dimension (D2) of a periphery of the second surface (**521**).

Clause 2.

[0080] The semiconductor element (**A10**) according to clause 1, wherein composition of the bonding layer (**60**) includes tin.

Clause 3.

[0081] The semiconductor element (**A10**) according to clause 2, [0082] further comprising a redistribution wiring (**40**) located between the electrode (**12**) and the terminal (**50**) in the first direction, wherein the redistribution wiring (**40**) electrically conducts to the electrode (**12**) and the terminal (**50**).

Clause 4.

[0083] The semiconductor element (**A10**) according to clause 3, wherein each of the first surface (**511**) and the electrode (**12**) is electrically connected to the redistribution wiring (**40**).

Clause 5.

[0084] The semiconductor element (**A10**) according to clause 4, [0085] further comprising a first protective film (**31**) covering the redistribution wiring (**40**), wherein [0086] the first protective film (**31**) is formed with a first opening (**311**) penetrating in the first direction and exposing the redistribution wiring (**40**), and [0087] a portion of the terminal (**50**) is housed in the first opening (**311**).

Clause 6.

[0088] The semiconductor element (**A10**) according to clause 5, wherein the terminal (**50**) includes a first portion (**51**) including the first surface (**511**) and housed in the first opening (**311**), and a second portion (**52**) including the second surface (**521**) and protruding from the first opening (**311**), and [0089] the second portion (**52**) is located outside the first opening (**311**) as viewed in the first direction.

Clause 7.

[0090] The semiconductor element (A10) according to clause 6, wherein the first portion (51) includes a first peripheral surface (512) surrounding the first surface (511) as viewed in the first direction, and [0091] the first peripheral surface (512) is in contact with the first protective film (31).

Clause 8.

[0092] The semiconductor element (A10) according to clause 7, wherein the first peripheral surface (512) is inclined with respect to the second direction, and [0093] a cross-sectional area of the first portion (51) in a direction orthogonal to the first direction decreases from the second portion (52) toward the first surface (511).

Clause 9.

[0094] The semiconductor element (A10) according to clause 8, wherein the second portion (52) includes a second peripheral surface (522) surrounding the second surface (521) as viewed in the first direction, and [0095] the first peripheral surface (512) is entirely located inwardly of the second peripheral surface (522) as viewed in the first direction.

Clause 10.

[0096] The semiconductor element (A10) according to clause 9, wherein the second peripheral surface (522) is inclined with respect to the second direction, and [0097] a cross-sectional area of the second portion (52) in a direction orthogonal to the first direction decreases from the second surface (521) toward the first portion (51).

Clause 11.

[0098] The semiconductor element (A20) according to clause 10, wherein a dimension (t2) of the second portion (52) in the first direction is greater than a dimension (t1) of the first portion (51) in the first direction.

Clause 12.

[0099] The semiconductor element (A20) according to clause 11, wherein the dimension (t2) of the second portion (52) in the first direction is equal to or greater than 5% of the dimension (D2) of the periphery of the second surface (521) in the second direction.

Clause 13.

[0100] The semiconductor element (A30) according to clause 12, further comprising an intermediate layer (70) located between the second surface (521) and the bonding layer (60), wherein [0101] the intermediate layer (70) contains a metal, and [0102] the intermediate layer (70) is in contact with each of the second surface (521) and the bonding layer (60).

Clause 14.

[0103] The semiconductor element (A30) according to clause 13, wherein composition of the intermediate layer (70) includes nickel.

Clause 15.

[0104] The semiconductor element (A10) according to any one of clauses 5 to 14, wherein the terminal (50) includes a first recess (53) recessed from the second surface (521), and [0105] a portion of the bonding layer (60) is housed in the first recess (53).

Clause 16.

[0106] The semiconductor element (A10) according to clause 15, further comprising a second protective film (32) located between the main body (11) and the redistribution wiring (40) in the first direction, wherein [0107] the second protective film (32) includes a second opening (321) penetrating in the first direction and exposing the electrode (12), and [0108] a portion of the redistribution wiring (40) is housed in the second opening (321).

Clause 17.

[0109] The semiconductor element (A10) according to clause 16, wherein the first recess (53) overlaps with the second opening (321) as viewed in the first direction.

Clause 18.

[0110] The semiconductor element (A40) according to clause 17, wherein the terminal (50)

includes a second recess (54) recessed in the first direction from a surface of the terminal (50) that defines the first recess (53), and [0111] a portion of the bonding layer (60) is housed in the second recess (54).

Clause 19.

[0112] A semiconductor device (B) comprising: [0113] the semiconductor element (A10) as set forth in clause 15, and [0114] a base (80) that includes a conductive part (82), wherein [0115] the semiconductor element (A10) is mounted on the base (80), and [0116] the terminal (50) is conductively bonded to the conductive part (82) via the bonding layer (60).

Clause 20.

[0117] The semiconductor element (A10) according to clause 16, wherein each of the first protective film (31) and the second protective film (32) contains polyimide.

Clause 21.

[0118] The semiconductor element (A10) according to clause 16, wherein a dimension of the first protective film (31) in the first direction is greater than the second dimension (32) of the second protective film.

Clause 22.

[0119] The semiconductor element (A10) according to clause 16, wherein the redistribution wiring (40) includes a main portion (41) located between the first protective film (31) and the second protective film (32), and a contact portion (42) connected to the main portion (41), [0120] the electrode (12) is electrically connected to the contact portion (42) and in contact with the second protective film (32), and [0121] the first surface (511) is electrically connected to the main portion (41).

Clause 23.

[0122] The semiconductor element (A10) according to clause 22, wherein the contact portion (42) is housed in the second opening (321).

Clause 24.

[0123] The semiconductor element (A10) according to clause 23, wherein the redistribution wiring (40) includes a dented portion (43) recessed from the main portion (41) in a first direction, [0124] the dented portion (43) overlaps with the contact portion (42) as viewed in the first direction, and [0125] the first protective film (31) is received in the dented portion (43).

Clause 25.

[0126] The semiconductor element (A40) according to clause 18, wherein the redistribution wiring (40) includes a dented portion (43) recessed from one side in a first direction, and [0127] a part of the first portion (51) is received in the dented portion (43).

Clause 26.

[0128] The semiconductor element (A40) according to clause 18, wherein the second recess (54) is located inwardly of the second opening (321) as viewed in the first direction.

Clause 27.

[0129] The semiconductor element (A50) according to clause 16, further comprising a third protective film (33) located opposite to the electrode (12) across the main body (11), [0130] wherein the third protective film (33) covers one side of the main body (11) in the first direction.

TABLE-US-00001 REFERENCE NUMERALS A10 to A50: Semiconductor element B:

Semiconductor device 11: Main body 11A: Obverse surface 111: Semiconductor substrate 112:

Semiconductor layer 12: Electrode 12A, 12B, 12C: First electrode, Second electrode, Third

electrode 20: Passivation film 21: Opening 31: First protective film 311: First opening 32: Second protective film 321: Second opening 33: Third protective film 40: Redistribution wiring 41: Main

portion 42: Contact portion 43: Dented portion 50: Terminal 51: First portion 511: First surface

512: First peripheral surface 52: Second portion 521: Second surface 522: Second peripheral

surface 53, 54: First recess, Second recess 60: Bonding layer 70: Intermediate layer 80: Base 81:

Substrate 82: Conductive part z, x, y: First direction, Second direction, Third direction

Claims

1. A semiconductor element comprising: a main body including a semiconductor layer; an electrode located on one side of the main body in a first direction and electrically conducting to the semiconductor layer; a terminal located opposite to the main body with respect to the electrode in the first direction and electrically conducting to the electrode; and a bonding layer electrically conducting to the terminal, wherein the bonding layer contains a metal, the terminal includes a first surface facing a side where the electrode is located in the first direction, and a second surface facing away from the first surface in the first direction, the bonding layer is located opposite to the first surface across the second surface, and in a second direction orthogonal to the first direction, a dimension of a periphery of the first surface is equal to or greater than 40% and equal to or less than 60% of a dimension of a periphery of the second surface.
2. The semiconductor element according to claim 1, wherein composition of the bonding layer includes tin.
3. The semiconductor element according to claim 2, further comprising a redistribution wiring located between the electrode and the terminal in the first direction, wherein the redistribution wiring electrically conducts to the electrode and the terminal.
4. The semiconductor element according to claim 3, wherein each of the first surface and the electrode is electrically connected to the redistribution wiring.
5. The semiconductor element according to claim 4, further comprising a first protective film covering the redistribution wiring, wherein the first protective film is formed with a first opening penetrating in the first direction and exposing the redistribution wiring, and a portion of the terminal is housed in the first opening.
6. The semiconductor element according to claim 5, wherein the terminal includes a first portion including the first surface and housed in the first opening, and a second portion including the second surface and protruding from the first opening, and the second portion is located outside the first opening as viewed in the first direction.
7. The semiconductor element according to claim 6, wherein the first portion includes a first peripheral surface surrounding the first surface as viewed in the first direction, and the first peripheral surface is in contact with the first protective film.
8. The semiconductor element according to claim 7, wherein the first peripheral surface is inclined with respect to the second direction, and a cross-sectional area of the first portion in a direction orthogonal to the first direction decreases from the second portion toward the first surface.
9. The semiconductor element according to claim 8, wherein the second portion includes a second peripheral surface surrounding the second surface as viewed in the first direction, and the first peripheral surface is entirely located inwardly of the second peripheral surface as viewed in the first direction.
10. The semiconductor element according to claim 9, wherein the second peripheral surface is inclined with respect to the second direction, and a cross-sectional area of the second portion in a direction orthogonal to the first direction decreases from the second surface toward the first portion.
11. The semiconductor element according to claim 10, wherein a dimension of the second portion in the first direction is greater than a dimension of the first portion in the first direction.
12. The semiconductor element according to claim 11, wherein the dimension of the second portion in the first direction is equal to or greater than 5% of the dimension of the periphery of the second surface in the second direction.
13. The semiconductor element according to claim 12, further comprising an intermediate layer located between the second surface and the bonding layer, wherein the intermediate layer contains a metal, and the intermediate layer is in contact with each of the second surface and the bonding

layer.

14. The semiconductor element according to claim 13, wherein composition of the intermediate layer includes nickel.

15. The semiconductor element according to claim 5, wherein the terminal includes a first recess recessed from the second surface, and a portion of the bonding layer is housed in the first recess.

16. The semiconductor element according to claim 15, further comprising a second protective film located between the main body and the redistribution wiring in the first direction, wherein the second protective film includes a second opening penetrating in the first direction and exposing the electrode, and a portion of the redistribution wiring is housed in the second opening.

17. The semiconductor element according to claim 16, wherein the first recess overlaps with the second opening as viewed in the first direction.

18. The semiconductor element according to claim 17, wherein the terminal includes a second recess recessed in the first direction from a surface of the terminal that defines the first recess, and a portion of the bonding layer is housed in the second recess.

19. A semiconductor device comprising: the semiconductor element as set forth in claim 15, and a base that includes a conductive part, wherein the semiconductor element is mounted on the base, and the terminal is conductively bonded to the conductive part via the bonding layer.
